SHIT Sciences Humanities Information Technology

构石

拖延的相变：截止压力下从连续拖延到突发赶工

A Phase-Transition Model of Procrastination Under Deadline Pressure

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ABSTRACT / 摘要 —Procrastination often exhibits a familiar qualitative pattern: long periods of negligible progress followed by a sudden "all-nighter" burst shortly before the deadline. We propose a minimal statistical-physics-inspired model in which task progress is an order parameter and deadline pressure acts as a time-dependent external field. Using a Landau free-energy landscape (Ising-like mean-field) with relaxational dynamics, we show that for sufficiently low self-control (i.e., a bistable landscape), increasing deadline pressure drives a saddle-node crossing (spinodal), producing an abrupt jump in the effective work rate. We provide "visual evidence" via (i) progress trajectories, (ii) a phase diagram in the plane of self-control vs. deadline intensity, and (iii) mean-field critical scaling of the spontaneous work-mode and the spinodal field with exponents 1/2 and 3/2. Although intentionally stylized, the model offers a compact language for describing bursty deadline behavior as a dynamical phase transition.

中文摘要: 拖延常呈现一种熟悉的质感: 大部分时间进度近乎停滞, 而在截止日前突然出现"通宵式"爆发。本文提出一个最小的、受统计物理启发的模型: 将任务进度视为序参量, 将截止压力视为随时间变化的外场。在 Landau 自由能 (类 Ising 平均场) 与弛豫动力学的框架下, 我们展示: 当自控参数足够低 (对应双稳势能面) 时, 随截止压力增强会触发鞍结分叉 (spinodal) 越过, 从而导致有效工作率的突跃。我们给出三类"视觉证据": (i) 进度随时间曲线; (ii) 自控参数与截止强度平面上的相图; (iii) 序参量与 spinodal 外场的平均场临界标度, 并得到指数 1/2 与 3/2。尽管模型刻意理想化, 它仍为"临近截止才爆发"的行为提供了一个可压缩、可作图、可拟合的相变语言。

INDEX TERMS / 关键词 Procrastination, deadline, phase transition, Ising model, Landau theory, hysteresis.

IMPACT STATEMENT / 影响声明 We turn "last-minute panic" into a controllable phase diagram: when and why a procrastinator flips from "sleep" to "sprint" can be framed as a spinodal transition driven by a deadline field.

I. INTRODUCTION / 引言

Deadlines are peculiar thermodynamic devices: they do nothing to the task itself, yet they reshape human behavior. Empirically, many people report "flat" progress for most of the available time and then a sharp acceleration near the deadline. The central question of this paper is deliberately narrow: *can the bursty switch from procrastination to work be captured as a phase transition / critical phenomenon?*

中文: 截止日期像一种奇怪的"热力学装置": 它并不改变任务本身, 却能显著重塑人的行为轨迹。经验上, 许多人会出现"长时间几乎不动、临近截止突然加速"的典型模式。本文故意把问题收窄: 这种从拖延到爆发式工作的切换, 能否用相变/临界现象来刻画?

Statistical physics offers a compact toolbox for describing abrupt qualitative change. In particular, the Ising/Landau picture organizes behavior around (i) an *order parameter* describing macroscopic state, (ii) a *control parameter* that tunes stability, and (iii) an *external field* that biases the state. Here we treat "working" vs. "not working" as competing basins in an effective landscape, while the deadline provides a time-dependent bias.

中文: 统计物理擅长描述"质变": 它用 (1) 描述宏观状态的序参量、(2) 调节稳定性的控制参量、以及 (3) 偏置状态的外场来组织现象。在这里, 我们把"工作/不工作"当作有效势能面中的两个竞争盆地, 而截止日期提供一个随时间增强的偏置外场。

Contributions. (1) We propose a minimal Landau (Ising-like mean-field) model for procrastination under a deadline. (2) We demonstrate a sharp "work-mode" jump via spinodal crossing in a time-dependent field. (3) We summarize the behavior in a phase diagram (deadline intensity vs. self-control) and report mean-field scaling exponents.

中文: 贡献: (1) 提出一个最小的 Landau (类 Ising 平均场) 拖延模型; (2) 在时间依赖外场下展示通过 spinodal 越过导致的"工作模式"突跃; (3) 用 (r, k) 相图总结行为, 并给出平均场标度指数。

II. MODEL / 模型

A. Order parameter and effective free energy

Let $m(t) \in [-\infty, \infty]$ denote a scalar *work-mode* order parameter. Intuitively, $m < 0$ corresponds to procrastination (low effort), while $m > 0$ corresponds to active work. We adopt a standard Landau potential

$$V(m; r, h) = \frac{r}{2}m^2 + \frac{1}{4}m^4 - hm, \tag{1}$$

where r is a "self-control" parameter and h is an external field representing deadline pressure. For $r > 0$ the potential is single-well; for $r < 0$ it becomes double-well and admits bistability (a procrastination basin and a work basin).

中文： 令 $m(t)$ 为标量工作模式序参量：$m < 0$ 代表拖延（低努力），$m > 0$ 代表投入工作。我们采用标准 Landau 势 $V(m; r, h)$（式(1)），其中 r 刻画"自控/自律"强度，h 为外场（截止压力）。$r > 0$ 时势能单井，对应平滑推进；$r < 0$ 时势能变为双井，允许拖延盆地与工作盆地共存（双稳）。

B. Deadline field schedule

We model a deadline at normalized time $t \in [0, 1]$. Deadline pressure ramps up as the remaining time $\Delta(t) = 1 - t$ shrinks. A convenient bounded schedule is

$$h(t) = k\, s(t), \qquad s(t) = \frac{\ln\left(\frac{1+\varepsilon}{1-t+\varepsilon}\right)}{\ln\left(\frac{1+\varepsilon}{\varepsilon}\right)} \in [0, 1], \tag{2}$$

where $k \geq 0$ is the deadline intensity and $\varepsilon > 0$ prevents divergence. Thus k is the *maximum* external field strength attained at the deadline.

中文： 我们把截止时刻归一化为 $t \in [0, 1]$。剩余时间 $\Delta(t) = 1 - t$ 越小，心理压力越大；因此选取一个有界递增的场强函数 $h(t)$（式(2)）。k 是截止强度（截止瞬间外场的最大值），ε 用于避免对数发散。

C. Dynamics and progress

We use relaxational dynamics with additive noise

$$\frac{dm}{dt} = -\mu \frac{\partial V}{\partial m} + \sqrt{2D}\,\eta(t) = \mu\,(h(t) - rm - m^3) + \sqrt{2D}\,\eta(t), \tag{3}$$

where μ is a response rate (mobility), D is a noise strength, and η is unit white noise.

Task progress $x(t) \in [0, 1]$ evolves as

$$\frac{dx}{dt} = v_{\max}\,\sigma(\gamma m), \qquad \sigma(u) = \frac{1}{1 + e^{-u}}, \tag{4}$$

with $v_{\max}$ the maximal "sprint" rate and γ a steepness parameter. Equation (4) simply maps work-mode to an effective work rate.

中文： 动力学采用弛豫形式（式(3)）：系统沿势能梯度下滑，同时叠加噪声模拟日常波动。进度 $x(t)$ 由 $m(t)$ 经过 Sigmoid 映射到工作率（式(4)），仅用于把"状态"转成"可观察的进度曲线"。

Model parameters used in figures

Parameter	Value	Meaning
ε	0.05	deadline smoothing in Eq. (2)
μ	12	response rate in Eq. (3)
D	0.002	noise strength
$v_{\max}$	9	maximal sprint rate in Eq. (4)
γ	4	sigmoid steepness

TABLE I: Default parameters for simulations (unless otherwise stated).

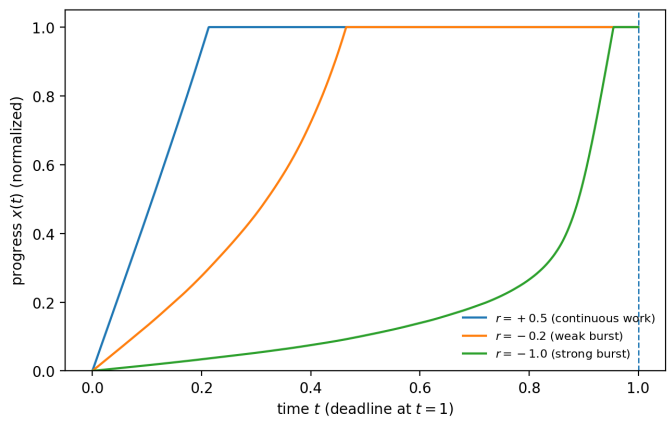

Fig. 1: Progress trajectories $x(t)$ under the same deadline intensity k but different self-control parameter r. The $r < 0$ cases display delayed activation and a sudden acceleration (burst) near the deadline.

III. RESULTS / 结果

A. From slow drift to sudden burst

Figure 1 shows representative progress trajectories. When $r > 0$ (higher self-control / single-well landscape), progress rises smoothly and saturates early. When $r < 0$, the system can remain trapped in the procrastination basin until the deadline field becomes strong enough to destabilize it, at which point $m(t)$ rapidly jumps upward and the progress rate spikes (a "burst").

中文： 图 1给出典型进度轨迹。$r > 0$（单井）时，进度随时间平滑增长并较早饱和；$r < 0$（双井）时，系统可能长期困在拖延盆地，直到外场足够强使拖延盆地失稳，此时 $m(t)$ 快速上跳、进度率陡增，呈现"临近截止突然爆发"。

B. Phase diagram in (r, k)

A key advantage of the phase-transition framing is that it organizes behavior in a compact phase diagram. We define "activation" as the existence of a time $t \leq 1$ such that $m(t)$ crosses from negative to positive (escape from the procrastination basin). Figure 2 maps this criterion across parameter space. For $r < 0$, activation requires sufficiently strong deadline intensity k. In the adiabatic limit (infinitely slow ramp and/or infinitely fast relaxation), the onset is controlled by the spinodal (saddle-node) condition

$$k_{\mathrm{sp}}(r) = \frac{2}{3\sqrt{3}}(-r)^{3/2}, \qquad (r < 0), \tag{5}$$

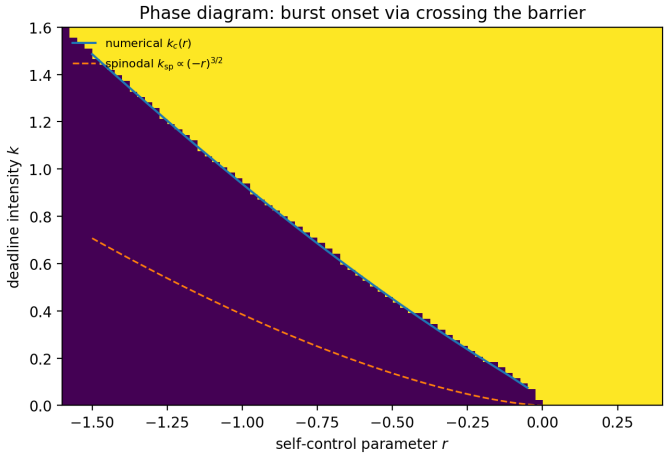

Fig. 2: Phase diagram (switch criterion) in the plane of self-control r and deadline intensity k. Colors indicate whether the system escapes the procrastination basin (m crosses to > 0) before the deadline. The dashed curve is the adiabatic spinodal prediction $k_{\rm sp} \propto (-r)^{3/2}$.

which we overlay for reference. Finite response rate shifts the numerical threshold upward because the system must cross the barrier *in time* before $t = 1$.

中文： 相变表述的直接收益，是可以把行为压缩成一张相图。我们把“激活”定义为：在 $t \le 1$ 内存在某时刻使 $m(t)$ 从负跨到正（逃离拖延盆地）。图 2展示该判据在 (r, k) 平面上的分区。$r < 0$ 时需要足够大的 k 才能激活。在绝热极限下，临界由 spinodal（鞍结分叉）给出（式(5)）。有限响应速度会把数值阈值推高，因为系统必须在截止前“来得及”翻过势垒。

C. *“Critical” scaling (mean-field exponents)*

Near the critical point $r \to 0^-$, the Landau model predicts standard mean-field scaling. At zero field, minimizing Eq. (1) gives a spontaneous work-mode magnitude

$$m_0 \sim (-r)^{1/2}, \qquad (r < 0), \tag{6}$$

and the spinodal field scales as Eq. (5), i.e. $k_{\rm sp} \sim (-r)^{3/2}$. Figure 3 confirms these exponents by log–log fitting.

中文： 在 $r \to 0^-$ 附近，Landau 模型给出标准平均场标度。零外场下最小化式(1)可得自发“工作模式”幅度 $m_0 \sim (-r)^{1/2}$；spinodal 外场由式(5)给出，因此 $k_{\rm sp} \sim (-r)^{3/2}$。图 3用 log–log 拟合验证了这两个指数。

IV. DISCUSSION / 讨论

Interpretation. In this model, “procrastination” is not moral failure but a metastable basin. A deadline increases an external bias $h(t)$; once the bias exceeds the spinodal, the procrastination basin disappears and the system must transition to the working state. This produces an abrupt increase in the work rate that resembles the subjective experience of last-minute panic.

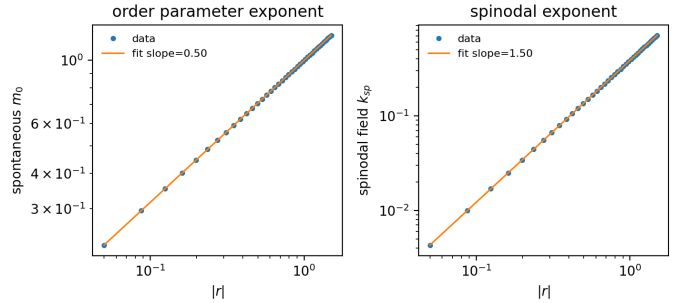

Fig. 3: Mean-field scaling in the Landau model: $m_0 \propto |r|^{1/2}$ and $k_{\rm sp} \propto |r|^{3/2}$. Fitted slopes match the expected exponents (“hardcore” enough for a deadline).

中文： 解释： 在该模型中，“拖延”不是道德评判，而是一个亚稳态盆地。截止日期通过外场 $h(t)$ 不断偏置系统；当外场超过 spinodal 后，拖延盆地消失，系统被迫跃迁到工作态，从而出现工作率的突增，类似主观体验中的“最后一刻恐慌”。

What is (not) claimed. We do not claim universal numerical exponents for real humans. The exponents here are mean-field artifacts of the chosen Landau polynomial. The value of the model is conceptual: it provides a language to separate (i) barrier height (self-control), (ii) field schedule (deadline policy), and (iii) response rate and noise (day-to-day variability).

中文： 不声称什么： 我们不声称真实人类行为存在普适临界指数；本文的指数来自平均场 Landau 多项式的形式选择。该模型的价值更偏概念化：把 (1) 势垒/自控强度、(2) 外场日程/截止制度、(3) 响应速度与噪声（随机波动）清晰分离。

Limitations and extensions. Obvious extensions include multi-task coupling (a many-spin model), heterogeneous “agents” (components of attention), explicit cost of effort, and adaptive deadlines. Empirically, the phase diagram could be calibrated using time-stamped activity logs (e.g., code commits), allowing one to estimate an effective r and k per individual.

中文： 局限与扩展： 可扩展方向包括：多任务耦合（多自旋模型）、注意力的多组分“代理”、显式努力成本，以及自适应/多段截止政策。经验上，可用带时间戳的活动日志（如 commit、写作记录）去标定个人的有效参数 r, k，得到个体化相图。

V. CONCLUSION / 结论

We framed deadline-driven procrastination as a dynamical phase-transition problem. A minimal Landau/Ising-like model with a time-dependent deadline field reproduces a common pattern: delayed progress followed by a sudden burst. The behavior is summarized by a phase diagram in (r, k) and exhibits mean-field scaling near the critical point. Even if the world refuses to be mean-field, the phase-transition metaphor remains a useful compression scheme for deadline behavior.

中文：我们将“截止驱动的拖延”表述为一个动力学相变问题。最小的 Landau/类 Ising 模型在时间依赖的截止外场下即可复现常见图景：长时间延迟推进，随后在截止前出现突发式爆发。相图 (r, k) 提供了紧凑总结，并在临界附近呈现平均场标度。即便真实世界并不遵循平均场，这种相变隐喻依然是描述与组织截止行为的一种有效压缩方式。

VI. MATERIALS AND METHODS / 材料与方法

All figures are produced by Euler–Maruyama simulation of Eq. (3) coupled to Eq. (4). A reference implementation (Python/NumPy/Matplotlib) is included as supplementary material in the project folder.

中文：所有图像均由对式(3)（与式(4)耦合）进行 Euler–Maruyama 数值模拟得到。项目文件夹中包含可复现的 Python/NumPy/Matplotlib 脚本。

SUPPLEMENTARY MATERIAL / 补充材料

(1) Script `make_figures.py`: generates all figures used in this manuscript.

(2) A short note on how the “deadline field” can be replaced by other ramp functions.

ACKNOWLEDGEMENT / 致谢

Thanks to the universal force of approaching deadlines.

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